

H2BENCH · DESIGN DECISION RECORD

H2Bench TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	HBN-DDR-001	Version	0.2
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/h2bench		
Source	docs/decisions/0001-trl2-review-decisions.md @ f294616		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review items adopted for TRL 3 work under Amish's 2026-09-25 instruction, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted for items 1 to 11, 13 and 14 (Amish, 2026-09-25: "i accept all your recommendations, go with them across all repos"; see HBN-DDR-002). Item 12 carried no recommendation and remains proposed, awaiting Amish.

■ Context

The TRL 2 review note (docs/REVIEW.md , session of 2026-09-25) listed nine H2Bench design choices as "Proposed, awaiting Amish", most with a recommendation. On 2026-09-25 Amish asked for this batch of Design Molecule repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed the H2Bench items one by one. This record therefore lists each recommendation as adopted for TRL 3 work, open for his review, and lists separately what stays open. In v0.1 nothing here was recorded as approved. Later on 2026-09-25 Amish accepted all the recommendations (HBN-DDR-002), and v0.2 records the items as decided. TRL 4 is on hold by Amish's instruction.

■ Options considered

The options for each item are in docs/REVIEW.md (TRL 2 session) and HBN-PRC-001. They are not repeated here.

■ Decision

Table 1. Items adopted for TRL 3 work, now decided (HBN-DDR-002).

#	Item	Status	Where it now lives
1	Storage: rigid 2 L buffer tank at up to 300 kPa gauge, pressurized by the electrolyzer; near-atmospheric gasbag with a pump stays the fallback if R10 cannot be met	Decided by Amish, 2026-09-25: go with recommendation	HBN-PRC-001 v0.3, <code>cad/src/model.py</code> , HBN-CAL-001 section 3
2	PEM electrolysis with deionized water, not alkaline	Decided by Amish, 2026-09-25: go with recommendation	HBN-PRC-001 v0.3
3	Generic 12 W class fuel cell with a published polarization curve as a purchase condition; branded stack as the fallback	Decided by Amish, 2026-09-25: go with recommendation	HBN-PRC-001 v0.3, <code>bom/bom.csv</code> line 12
4	Hydrogen measured by pressure, volume and temperature, not by a mass flow meter	Decided by Amish, 2026-09-25: go with recommendation	HBN-PRC-001 v0.3, HBN-CAL-001 section 6
5	Bench power supply in constant-current mode as the source; solar as a later option	Decided by Amish, 2026-09-25: go with recommendation	HBN-PRC-001 v0.3
6	H2Guard costed in its own project but shipped with H2Bench; the bench cannot run without it	Decided by Amish, 2026-09-25: go with recommendation	HBN-REQ-001 v0.3 R14, <code>bom/bom.csv</code> line 15
7	Tank kept above about 20 kPa gauge between lessons; a new or opened tank purged by three pressure cycles	Decided by Amish, 2026-09-25: go with recommendation	HBN-PRC-001 v0.3, HBN-PRB-001 v0.3
8	First users: vocational hydrogen technician training, ahead of secondary school practicals	Decided by Amish, 2026-09-25: go with recommendation	HBN-PRB-001 v0.3

No pitch or problem rewording was recommended at TRL 2, so `project.yaml` and `README.md` keep their pitch and problem lines.

Items listed as open in v0.1

Table 2. Items open in v0.1, with their status after HBN-DDR-002.

#	Item	Status
9	Budget. The TRL 2 review recommended raising <code>budget_usd</code> from \$450 to about \$850 (bench power supply included, H2Guard costed separately). <code>budget_usd</code> stays at \$450 in <code>project.yaml</code> . HBN-CAL-001 prices the kit at \$865 in full, \$800 without the bench supply and \$780 without the supply and the plug-in RCD	Decided by Amish, 2026-09-25: go with recommendation: <code>budget_usd</code> raised from \$450 to \$850 (HBN-DDR-002 D9)
10	Whether the bench power supply is part of the kit or supplied by the school (BOM line 4)	Decided by Amish, 2026-09-25: go with recommendation: the supply is in the kit, as recommendation (a) of item 9 stated (HBN-DDR-002 D10)
11	Canopy hood with a single high point for sensing and extraction (HBN-PRC-001 choice 8). It was not in the TRL 2 review list	Decided by Amish, 2026-09-25: go with recommendation: the canopy hood proposed as the working design is kept (HBN-DDR-002 D8)
12	Curriculum to align the first worksheets with; class demonstration or group practical first; aluminium or stainless tank	Proposed, awaiting Amish. No recommendation was made
13	Engineering proposals from HBN-CAL-001: relief set at 325 kPa gauge (329 kPa gauge or less keeps R5 at full relief lift); a 310 kPa gauge high-pressure cut of the supply so the relief valve is only a backup; vent needle valve set for 3 min or more; plug-in 30 mA RCD as BOM line 18	Decided by Amish, 2026-09-25: go with recommendation: relief 325 kPa gauge, 310 kPa gauge supply cut, needle valve 3 min or more, RCD line (HBN-DDR-002 D11 to D14)
14	H2Guard interface: H2Guard's valve is 24 V, normally closed; H2Bench's tank solenoid is 12 V. H2Guard's first-user recommendation (university teaching lab) differs from item 8 here	Decided by Amish, 2026-09-25: go with recommendation: settle the interface with H2Guard; listed under cross-repo actions in <code>docs/REVIEW.md</code> ; H2Guard not edited

■ Consequences

- The design at TRL 3 is the rigid-tank, PEM, generic-stack bench measured by the gas law, as modeled in `cad/src/model.py` and drawn on HBN-DWG-001.
- Cost stays **not met**: after HBN-DDR-002 the kit is \$885 against the \$850 in `project.yaml`.
- R10 (electrolyzer back-pressure) remains the condition for item 1; if no generic stack can be shown to hold 332 kPa over its oxygen side (357 kPa before the relief moved to 325 kPa gauge), the fallback in item 1 applies.
- The first-user choice (item 8) makes supervision by trained instructors the planning assumption; the worksheets for school use wait for item 12.